BENDING STRESS CALCULATION FOR ARM

Given:

$m = 1.5$ kg

$g = 9.81\ m/s^2$

$W = m \times g = 1.5 \times 9.81 = 14.7$ N

Number of wheels = 3

Arm length $L = 90$ mm

Thickness $t = 3.5$ mm

Width $b = 20$ mm

Step 1: Load per wheel

$F = W / 3 = 14.7 / 3 = 4.9$ N

Step 2: Bending moment

$M = F \times L = 4.9 \times 90 = 441$ N·mm

Step 3: Moment of inertia

$I = (b \times t^3) / 12$

$I = (20 \times 3.5^3) / 12$

$I = (20 \times 42.875) / 12 = 71.46\ mm^4$

Step 4: Distance from neutral axis

$c = t / 2 = 3.5 / 2 = 1.75\ mm$

Step 5: Bending stress

$\sigma = (M \times c) / I$

$\sigma = (441 \times 1.75) / 71.46$

$\sigma \approx 10.8\ MPa$

Final:

$\sigma \approx 11\ MPa$